FA 5 – LINEAR MODELS

Palmer Penguins Dataset: One-Way ANOVA, Dummy-Variable Regression, Two-Way ANOVA, and Interaction Effects

Research question

Does penguin body mass differ according to species and sex, and does the effect of species on body mass depend on sex?

Dataset and variables

The analysis uses the Palmer Penguins dataset. Body mass (grams) is the response variable; species (Adelie, Chinstrap, Gentoo) and sex (female, male) are categorical explanatory variables. The raw data contain 344 observations. Eleven observations were removed because at least one of body_mass_g, species, or sex was missing, leaving n = 333 complete observations.

A. Data preparation

The data were imported from data/penguins.csv, inspected using structure and summary output, and checked for missing values. Rows missing body_mass_g, species, or sex were excluded. Species and sex were converted to factors. The cleaned sample size is 333.

B. Exploratory data analysis

Statistic	Value
n	333
Mean body mass	4,207.06 g
Standard deviation	805.22 g
Minimum	2,700 g
Maximum	6,300 g

By species

Species	n	Mean (g)	SD (g)
Adelie	146	3706.16	458.62
Chinstrap	68	3733.09	384.34
Gentoo	119	5092.44	501.48

By sex

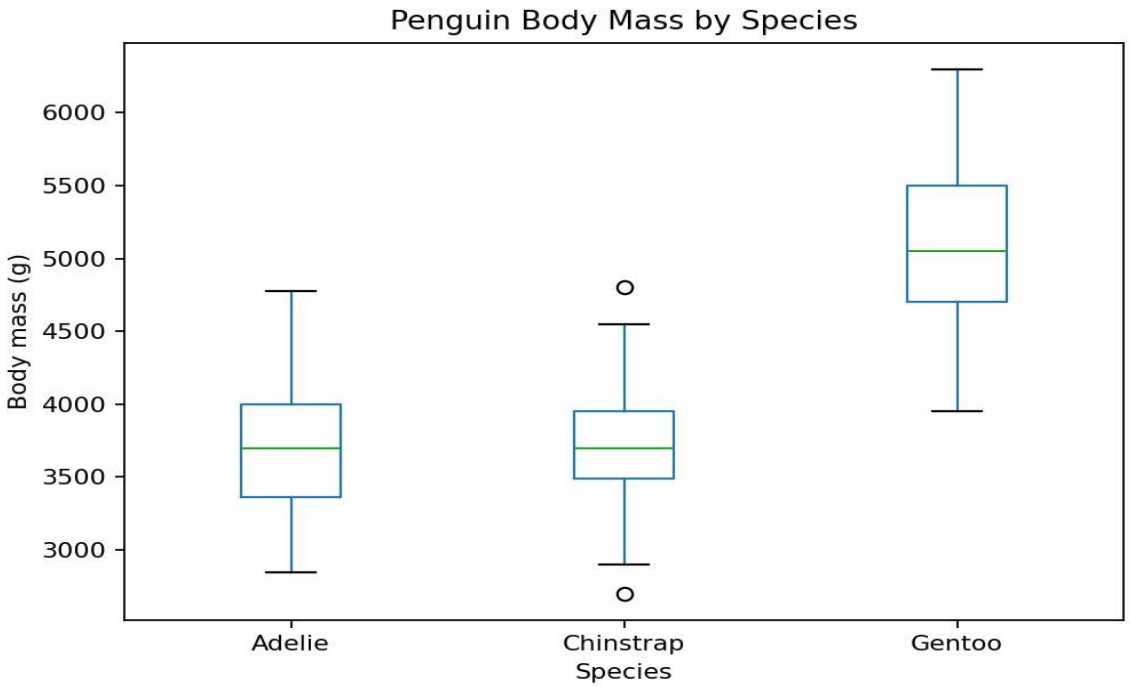
Sex	n	Mean (g)	SD (g)
female	165	3862.27	666.17
male	168	4545.68	787.63

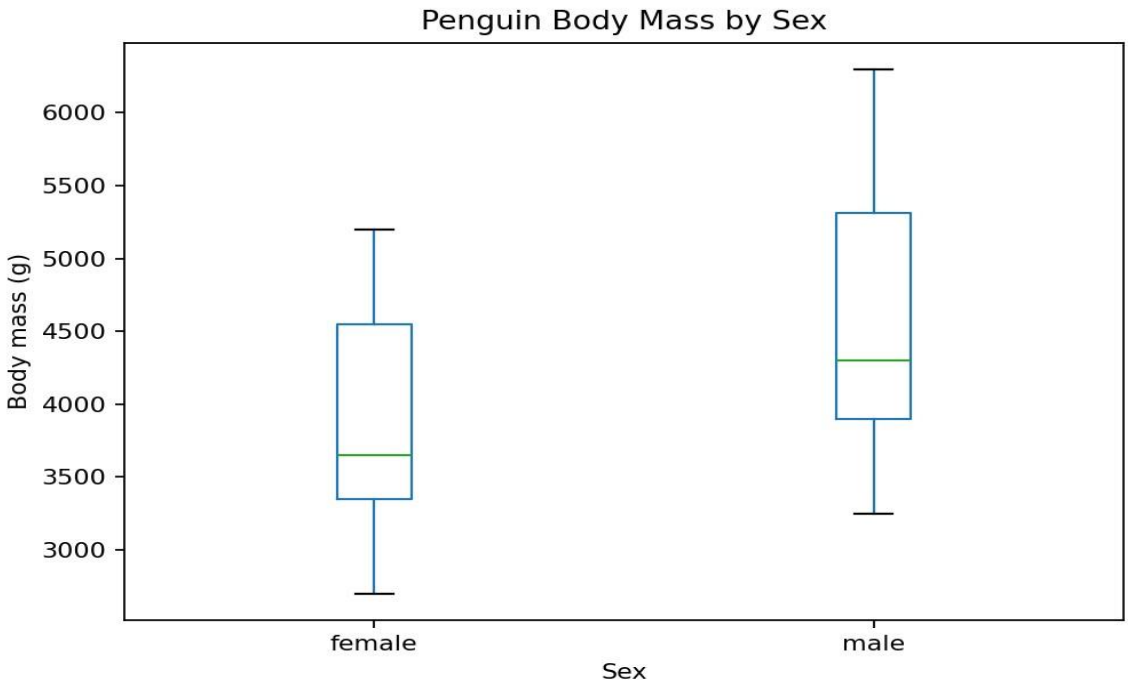
By species × sex

Species	Sex	n	Mean (g)	SD (g)
Adelie	female	73	3368.84	269.38
Adelie	male	73	4043.49	346.81

Chinstrap	female	34	3527.21	285.33
Chinstrap	male	34	3938.97	362.14
Gentoo	female	58	4679.74	281.58
Gentoo	male	61	5484.84	313.16

The Gentoo penguins have the highest average body mass. Males have a higher mean body mass than females overall. The male–female difference varies across species, suggesting a possible interaction; this is tested formally below.




Penguin Body Mass by Sex
Body mass (g)
6000
5500
5000
4500
4000
3500
3000
female
male
Sex

C. One-Way ANOVA

Hypotheses: H0: the mean body mass is equal for all three species (μAdelie = μChinstrap = μGentoo). H1: at least one species mean differs.

Source	df	Sum of squares	Mean square	F	p-value
Species	2	145190219.11	72595109.56	341.895	3.74 × 10^-81
Residual	330	70069446.80	212331.66	—	—

Decision at α = 0.05: reject H0 because p < 0.05. There is very strong statistical evidence that mean body mass differs among species. The sample means show Gentoo penguins are much heavier on average than Adelie and Chinstrap penguins. The ANOVA establishes that at least one mean differs; it does not, by itself, prove every pair differs.

D. Dummy-variable regression

Using Adelie as the reference category, the fitted model is: body_mass_g = β0 + β1D(Chinstrap) + β2D(Gentoo) + ε.

Term	Estimate (g)	Interpretation
Intercept	3706.16	Estimated mean for Adelie.
Chinstrap	26.92	Chinstrap mean minus Adelie mean.
Gentoo	1386.27	Gentoo mean minus Adelie mean.

The intercept estimates Adelie mean body mass at 3706.16 g. Chinstrap penguins average 26.92 g more than Adelie; this difference is not statistically significant (p = 0.691). Gentoo penguins average 1386.27 g more than Adelie, a statistically significant difference (p < 0.001). This regression is algebraically equivalent to the one-way ANOVA for the species factor.

E. Two-Way ANOVA

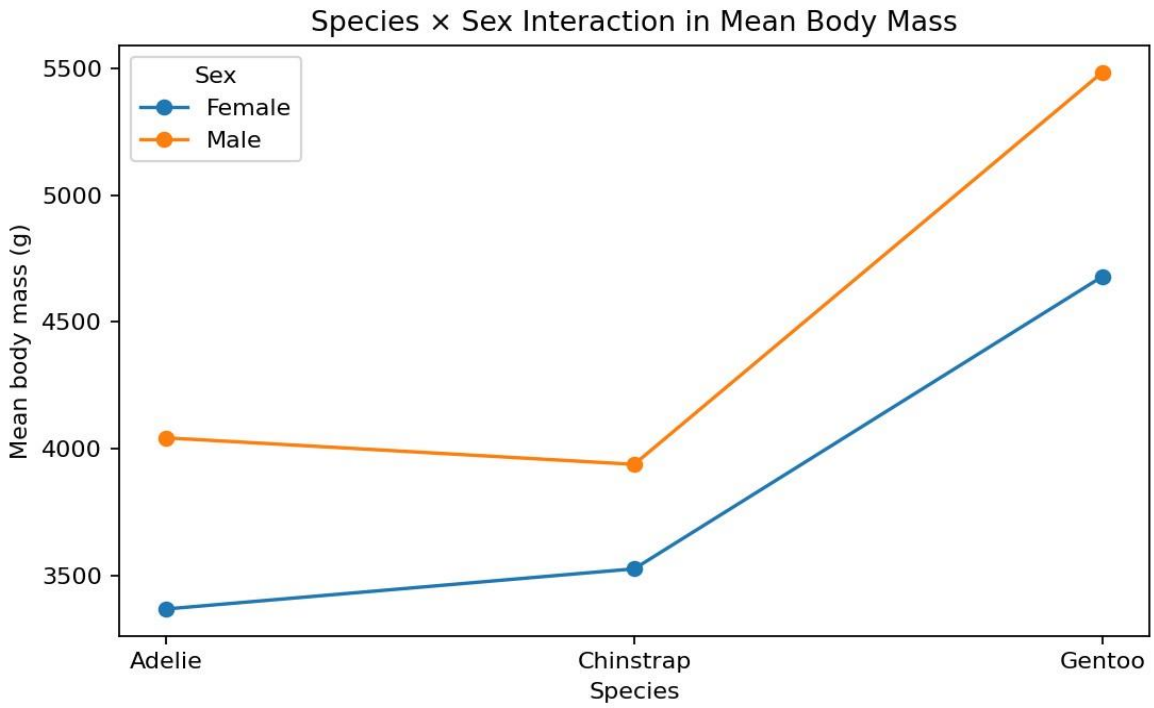
Model: body_mass_g ~ species + sex + species × sex. Each main effect is interpreted alongside the interaction term. The table below uses sequential (Type I) sums of squares, matching the default order of terms in R's aov() model.

Effect	df	Sum of squares	Mean square	F	p-value
Species	2	145190219.11	72595109.56	758.358	< 0.001
Sex	1	37090261.78	37090261.78	387.460	< 0.001
Species × Sex	2	1676556.74	838278.37	8.757	< 0.001
Residual	327	31302628.28	95726.69	—	—

Species main effect: significant (F(2,327) = 758.358, p < 0.001). Sex main effect: significant (F(1,327) = 387.460, p < 0.001). Species × sex interaction: significant (F(2,327) = 8.757, p = 0.000197). Thus, average body mass differs by species and sex, and the size of the sex difference is not the same for all species.

F. Interaction analysis

The interaction plot shows mean body mass for females and males across species. The lines are not parallel: the male–female difference is approximately 674.66 g for Adelie, 411.76 g for Chinstrap, and 805.09 g for Gentoo. This visual pattern is consistent with the statistically significant Species × Sex interaction.



Practical interpretation: males are heavier than females in all three species in this sample, but the size of that difference varies. The difference is smallest among Chinstrap penguins and largest among Gentoo penguins. Because the interaction is significant, main effects should not be interpreted as if the sex difference were identical across species.

G. Full regression model

With Adelie and female as reference categories, the fitted model is:

Body mass = β0 + β1D(Chinstrap) + β2D(Gentoo) + β3D(Male) + β4D(Chinstrap × Male) + β5D(Gentoo × Male) + ε.

Coefficient	Estimate (g)
Intercept	3368.84
species[T.Chinstrap]	158.37
species[T.Gentoo]	1310.91
sex[T.male]	674.66
species[T.Chinstrap]:sex[T.male]	-262.89
species[T.Gentoo]:sex[T.male]	130.44

The intercept (3368.84 g) is the estimated mean for female Adelie penguins. The male coefficient (674.66 g) is the male–female difference for Adelie. The Chinstrap × Male coefficient (-262.89 g) is the change in that sex difference for Chinstrap relative to Adelie; it is negative, indicating a smaller sex gap. The Gentoo × Male coefficient (130.44 g) indicates a larger sex gap than Adelie, although this coefficient is not statistically significant at 0.05 ($p = 0.089$). The overall interaction test is significant.

Final interpretation

The data provide strong evidence that body mass differs by species and sex. Gentoo penguins have the highest mean body mass, and males have higher mean body mass than females in each species. The species-by-sex interaction is statistically significant, so the male–female difference depends on species rather than remaining constant across all species. These are associations in the observed sample; the analysis does not establish that species or sex causally produces the differences.